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Structural Limits of State-Driven Adaptive Nonlocal Kernels in Galaxy Dynamics

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ABSTRACT

Context. Nonlocal gravity models with fixed kernel scales provide alternative descriptions of galaxy rotation curves but lack mechanisms for local, regime-dependent structural coupling.
Aims. We characterize the operational limits of state-driven adaptive kernel deformation under canonical baseline constraints.
Methods. We introduce an adaptive framework (H2) that modulates the effective kernel scale $L_{\rm eff}(r)$ via a local baryonic gradient field χ, while preserving the global baseline parameters (L_0 = 200 kpc, μ = 10). We apply a diagnostic pipeline to the canonical-baseline-compatible subset of 80 SPARC galaxies spanning baryon-dominated to dark-matter-dominated regimes (39 baryon-dominated, 30 balanced, 11 DM-dominated; $V_{\rm obs}/V_b$ = 0.027 to 2.187), quantifying changes in rotation curve scatter and state-residual correlation relative to the frozen baseline.
Results. We identify three key findings: (1) The Pearson correlation between χ and velocity residuals ΔV varies systematically across dynamical regimes (r = −0.371 to +0.981), supporting a regime-dependent adaptive response. (2) The empirical operating floor remains fixed at $L_0/3 \approx 66.67$ kpc, and all systems reach or approach this lower bound ($L_{\rm eff,min}$ range: 66.67–96.95 kpc), establishing a structural boundary within the validated fleet. (3) $\Delta\sigma$ is consistent with zero for all galaxies with sufficient inner-region coverage; 6 systems are undefined due to inner-coverage limitations. This is consistent with suppression by quadrature coupling, together with the empirical operating floor and the bounded adaptive range of the present implementation.
Conclusions. Local state-driven adaptation under fixed baseline scaling produces coherent regime-dependent coupling but is geometrically bounded and dynamically neutral with respect to empirical scatter. The framework provides a reproducible diagnostic benchmark for testing nonlocal kernel extensions and identifies specific structural limits that constrain the design space of adaptive gravitational models.

Key words. galaxies: kinematics and dynamics – gravitation – methods: numerical – methods: data analysis

1. Introduction

1.1. Empirical rotation curve problem

Observed galaxy rotation curves exhibit persistent discrepancies between the circular velocities inferred from kinematic measurements and those predicted from the distribution of luminous matter alone. Since the pioneering studies of spiral galaxies by Rubin et al. (1980) and Bosma (1981), it has been established that rotation curves remain approximately flat or slowly rising at large radii, where the baryonic surface density declines. This behavior is commonly interpreted either as evidence for extended dark matter halos (Navarro et al. 1997) or as a modification of gravitational dynamics at low accelerations (Milgrom 1983).

High-quality kinematic and photometric datasets, such as the SPARC sample (Lelli et al. 2016), provide a stringent empirical benchmark for testing mass models across a wide range of galaxy morphologies and baryonic fractions. These data reveal not only the large-scale mass discrepancy, but also systematic correlations between baryonic acceleration and total acceleration (McGaugh et al. 2016; Lelli et al. 2017), as well as residual structures in certain dynamical regimes. Any phenomenological framework must therefore be evaluated not only by its ability to reproduce global rotation curve amplitudes, but also by its behavior across baryon-dominated and dark-matter-dominated regions.

1.2. Nonlocal kernel framework

One phenomenological approach to modeling galactic dynamics consists of representing the gravitational response as a convolution of the baryonic density field with a nonlocal kernel. In this formulation, the effective circular velocity is determined by the baryonic mass distribution modulated through a scale-dependent kernel that encodes extended gravitational coupling. Fixed-scale nonlocal models can reproduce a broad range of rotation curve profiles with a limited number of global parameters.

In a previous implementation (hereafter H1), a frozen, isotropic kernel with a single effective scale was applied uniformly across a sample of disk galaxies. While this baseline configuration achieved reasonable agreement with observed velocity amplitudes in many systems, diagnostic analysis revealed structured residuals that depend on the local dynamical regime. In particular, baryon-dominated inner regions exhibited systematic overshoot, and correlation patterns indicated incomplete coupling between local baryonic structure and the effective gravitational response.

1.3. Nonlocal gravity approaches

Nonlocal modifications of gravity have been explored both theoretically and phenomenologically. Theoretical frame-

works include Mashhoon's nonlocal gravity (Mashhoon 2008; Hehl & Mashhoon 2009), cosmological nonlocality (Deser & Woodard 2007; Deffayet & Woodard 2009), and retarded gravitational fields (Chicone & Mashhoon 2016).

For rotation curve modeling specifically, Li et al. (2018) performed systematic SPARC analysis with multiple mass models, Rodrigues et al. (2018) tested the radial acceleration relation, and Lelli et al. (2017) characterized universal scaling behavior. MOND frameworks (Milgrom 1983; Famaey & McGaugh 2012) and NFW dark matter profiles (Navarro et al. 1997) provide standard benchmarks.

The present work differs by isolating the specific question of whether bounded local kernel adaptation can reduce systematic residuals in a fixed-baseline phenomenological framework.

1.4. Identified limitations

The presence of regime-dependent residual structure suggests that a single-scale, globally fixed kernel may not fully capture variations in local baryonic configuration. In the H1 baseline, deviations were most pronounced in high-surface-density systems and in transition regions between baryon-dominated and low-acceleration regimes. These features motivate testing whether allowing limited local adaptation of the kernel scale can reduce systematic bias without introducing unconstrained degrees of freedom.

Importantly, the goal of such an extension is *not* to construct a competing cosmological model, nor to eliminate the need for dark matter, but rather to assess the structural limits of local kernel adaptation under fixed global constraints. By isolating the effect of scale modulation, one can determine whether residual patterns originate from insufficient dimensionality in the kernel representation or from more fundamental modeling assumptions.

1.5. Scope and aims

In this work, we introduce H2, an adaptive extension of the frozen nonlocal kernel framework in which the effective kernel scale responds to local baryonic structure while preserving the global normalization and quadrature combination used in the baseline model. The adaptive mechanism is designed to remain bounded and diagnostically transparent, enabling a direct comparison with the H1 configuration.

The primary objective is to quantify whether state-dependent scale modulation modifies empirical scatter, correlation structure, and regime-dependent behavior in a validation sample of 80 SPARC galaxies spanning baryon-dominated, balanced, and dark-matter-dominated dynamical regimes (39 baryon-dominated, 30 balanced, 11 DM-dominated). Particular attention is given to identifying empirical operating floors and assessing whether local adaptation alone is sufficient to alter the inner scatter metric.

The H2 framework and associated diagnostic pipeline are publicly archived (Figueiredo 2026). This paper is organized as follows. Section 2 summarizes the baseline framework and introduces the adaptive kernel formulation. Section 3 describes the numerical implementation and data selection. Section 4 presents the results across dynamical regimes. Section 5 discusses the structural implications and limitations of local adaptation. Conclusions are given in Section 6.

2. Framework description

2.1. H1 baseline recap

The H1 baseline model represents the effective circular velocity as a quadrature combination of the Newtonian baryonic contribution and a nonlocal correction term derived from convolution with a fixed-scale kernel. The total model velocity is written as

$$V_{\rm tot}^2(r) = V_{\rm b}^2(r) + V_{\rm NL}^2(r), \tag{1}$$

where $V_{\rm b}(r)$ is the baryonic circular velocity computed from the observed mass distribution, and $V_{\rm NL}(r)$ is the nonlocal contribution. The nonlocal term is obtained by convolving the baryonic surface density $\Sigma_{\rm b}(r)$ with an isotropic kernel $K(r; L_0)$ characterized by a single global scale L_0, such that

$$V_{\rm NL}(r) = \mu\,\mathcal{F}\left[\Sigma_{\rm b}(r), K(r; L_0)\right], \tag{2}$$

where μ is a dimensionless normalization parameter and $\mathcal{F}$ denotes the convolution operator projected into circular velocity space. In the frozen H1 configuration, both L_0 and μ are treated as global constants across the galaxy sample. This formulation preserves the Newtonian baryonic structure while introducing a nonlocal modification governed by a single effective length scale. Diagnostic analysis of H1 revealed regime-dependent residual structure, motivating the adaptive extension described below. The archived implementation used in this work, including the explicit kernel form, taper prescription, and amplitude correction, is summarized in Section 3.3 and documented in full in Appendix A.

2.2. Adaptive extension mechanism

To assess whether limited local adaptation can reduce regime-dependent residual structure, we introduce a scale-modulation field $\chi(r)$, defined as a dimensionless measure of baryonic gradient stiffness,

$$\chi(r) = \left|\frac{{\rm d}\ln\Sigma_{\rm b}(r)}{{\rm d}\ln r}\right|. \tag{3}$$

The adaptive model modifies the effective kernel scale according to

$$L_{\rm eff}(r) = L_0\, f(\chi(r)), \tag{4}$$

where the response function takes the explicit form

$$f(\chi) = \frac{1}{1+\alpha\chi}, \tag{5}$$

with $\alpha = 2.0$ fixed throughout. The value $\alpha = 2.0$ was adopted as a fixed implementation parameter providing a bounded, monotonic response across the observed stiffness range ($0 \le \chi \lesssim 1$), while retaining sensitivity to inner-region baryonic gradients; it is not fitted to individual galaxies. Over the 80-galaxy sample, the adaptive formula yields the empirical operating range

$$\frac{L_0}{3} \le L_{\rm eff}(r) \le L_0, \tag{6}$$

corresponding to stiffness values $0 \le \chi \lesssim 1$ encountered in high-stiffness inner regions. The upper bound recovers the frozen H1 configuration in the limit of weak modulation ($\chi \to 0$). Critically, the normalization parameter μ remains fixed at its H1 value, ensuring that any changes in model behavior arise solely from scale adaptation rather than amplitude re-optimization.

2.3. Discretized kernel basis

Rather than allowing continuous variation of $L_{\rm eff}(r)$, the adaptive framework employs a discretized kernel basis consisting of eight pre-computed kernel tables at fixed scale multipliers

$$m_i \in \{0.05,\ 0.15,\ 0.25,\ 0.40,\ 0.55,\ 0.70,\ 0.85,\ 1.00\}, \tag{7}$$

corresponding to basis scales $L_i = m_i L_0 \in \{10, 30, 50, 80, 110, 140, 170, 200\}$ kpc at $L_0 = 200$ kpc. At each radius, $L_{\rm eff}(r)$ is mapped to the nearest basis element L_i, and the corresponding kernel table is applied. This discretization ensures numerical stability within the evaluation pipeline, which was implemented using standard scientific Python libraries (Harris et al. 2020; Virtanen et al. 2020), and limits effective model dimensionality.

The empirical lower bound at $L_0/3 = 66.67$ kpc (observed consistently across the 80-galaxy sample) reflects the typical stiffness ceiling $\chi \lesssim 1$ reached in high-stiffness inner regions under the taper configuration (Eq. 5), rather than the hard minimum of the basis ($m_{\rm min} = 0.05$). As shown in subsequent sections, this empirical floor plays a central role in determining whether local scale modulation can alter the inner scatter metric under quadrature combination.

2.4. Diagnostic protocol

The adaptive framework is evaluated through three targeted tests.

Test 1: Outer-regime stability. We verify that scale modulation does not introduce spurious deviations in low-acceleration outer regions where the H1 baseline performs consistently.

Test 2: χ–ΔV correlation. We quantify the Pearson correlation between the stiffness field $\chi(r)$ and velocity residuals $\Delta V(r) = V_{\rm H2}(r) - V_{\rm H1}(r)$ to assess whether the adaptive mechanism responds coherently to local baryonic structure.

Test 3: Inner-regime scatter neutrality. We compute the change in the inner scatter metric,

$$\Delta\sigma = \sigma_{\rm H2} - \sigma_{\rm H1}, \tag{8}$$

across the selected galaxy sample. Particular attention is given to whether scale modulation produces a statistically measurable reduction in inner-regime residuals.

All diagnostics are performed using identical data, radial sampling, and numerical tolerances as in the H1 baseline to ensure controlled comparability.

3. Numerical implementation and data

3.1. Galaxy sample

The adaptive framework is evaluated using the canonical-baseline-compatible subset of the SPARC database (Lelli et al. 2016), which provides high-quality rotation curves and homogeneous photometric decompositions. The H2 validation fleet comprises 80 SPARC galaxies satisfying the H2 compatibility condition $(L_0, \mu) = (200\,\text{kpc},\ 10)$, spanning the full range of dynamical regimes from extreme baryon domination to dark-matter domination:

- 39 baryon-dominated systems ($V_{\rm obs}/V_b < 0.8$)
- 30 balanced systems ($0.8 \le V_{\rm obs}/V_b < 1.5$)
- 11 dark-matter-dominated systems ($V_{\rm obs}/V_b \ge 1.5$)

These regime labels refer to the adopted fleet-level proxy $V_{\rm obs}/V_b$ and are used only as empirical classification bins; they are not intended as point-by-point statements of local mass dominance at every radius. The fleet was constructed incrementally: 9 original baseline galaxies (NGC0891, NGC6946, NGC5585, IC2574, NGC3893, NGC2903, NGC5055, NGC3198, NGC6503), plus 21 additional galaxies (Phase 1), plus 20 galaxies (Phase 2a), plus 30 galaxies (Phase 2b), for a total of 80 systems.

All 80 galaxies share the frozen H1 baseline parameters $L_0 = 200$ kpc and $\mu = 10$, defining the compatibility condition for H2 adaptive convolution under taper constraints. Galaxies with different H1 best-fit parameters require recalibration of taper settings and were excluded to preserve homogeneous adaptive conditions across the sample and to avoid introducing per-subset recalibration of the frozen H1 baseline. Across the 80-galaxy validation sample, the H1 baseline exhibits a median MAFE of 0.21, with a range of 0.07 to 1.67, spanning from well-constrained fits to highly uncertain baselines.

Six galaxies (F567-2, UGC09992, UGC05999, NGC6789, NGC4389, and UGC00634) showed insufficient radial data points in the inner region for scatter computation (data geometry limitation), yielding $\Delta\sigma$ = N/A. These systems are retained in the validation sample because they demonstrate adaptive kernel response (with Pearson r values comparable to measurable systems), but they are excluded from scatter statistics. The complete per-galaxy diagnostic results are provided in the archived repository as `fleet_summary_80galaxy.csv`.

3.2. Baryonic mass modeling

The baryonic circular velocity $V_{\rm b}(r)$ is computed from the observed stellar and gas surface density profiles provided in the SPARC database. Stellar mass-to-light ratios are adopted from the standard SPARC photometric analysis, and gas contributions follow the same assumptions adopted in the H1 baseline configuration. No additional free parameters are introduced in the adaptive extension.

Radial sampling, interpolation methods, and unit conventions are identical to those used in H1 to avoid numerical inconsistencies between baseline and adaptive runs.

3.3. Kernel evaluation and discretization

The nonlocal convolution operator $\mathcal{F}$ (Eq. 2) employs a regularized logarithmic kernel

$$U(r;\, L) = \tfrac{1}{2}\ln\!\left(1 + \frac{r^2 + (0.01\,L)^2}{L^2}\right), \tag{9}$$

where the small offset $(0.01\,L)^2$ regularizes the $r \to 0$ behaviour. A finite-support radial taper blends $L_{\rm eff}(r)$ smoothly toward L_0 at the outer boundary of each galaxy profile, ensuring that the frozen H1 configuration is recovered at large radii. A global amplitude correction factor $\beta = 1.15$ is applied consistently across all galaxies and is held fixed throughout H2. This factor compensates for the amplitude offset between the discrete grid evaluation and the target continuous kernel normalization; it was established as a fixed implementation constant prior to the validation runs and is not adjusted per galaxy.

The operator $\mathcal{F}$ is implemented numerically using direct radial integration with pre-computed kernel tables corresponding to the discrete scale set $m_i L_0$ (Eq. 7). For each radial grid point, $L_{\rm eff}(r)$ is mapped to the nearest discrete basis element and the corresponding kernel table is applied.

All computations were performed using standard scientific Python libraries (Harris et al. 2020; Virtanen et al. 2020). Visualization and data analysis utilized additional tools from the scientific ecosystem (Hunter 2007; McKinney 2010). Numerical precision tolerances and integration resolution were matched to those of the H1 baseline to ensure consistency. No re-optimization of μ or L_0 was performed in H2.

3.4. Residual definitions and statistical metrics

Velocity residuals are defined relative to the frozen H1 baseline as

$$\Delta V(r) = V_{\rm H2}(r) - V_{\rm H1}(r). \tag{10}$$

The scatter metric is computed as the root-mean-square deviation between model and observed velocities,

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(V_{\rm model}(r_i) - V_{\rm obs}(r_i)\right)^2}, \tag{11}$$

where N is the number of radial data points in the evaluated sample. All scatter values reported for H2 use identical radial masks and galaxy weighting as in H1. In the present diagnostic framework, σ is evaluated as an inner-region scatter metric wherever sufficient radial coverage is available.

Pearson correlation coefficients between $\chi(r)$ and $\Delta V(r)$ are computed over the same radial domain to assess adaptive coherence.

3.5. Reproducibility

The full H2 implementation, including kernel tables, galaxy selection criteria, and diagnostic scripts, is publicly archived under a versioned release (Figueiredo 2026). All numerical results presented in this work can be reproduced using the provided configuration files without parameter retuning. Complete documentation of the diagnostic pipeline is provided in the repository.

4. Results

4.1. Inner-region scatter neutrality

The primary diagnostic for the adaptive framework is the change in the inner-region scatter metric relative to the H1 baseline. Across the 80-galaxy validation sample, $\Delta\sigma$ is numerically consistent with zero for all galaxies with sufficient inner-region coverage (74/80), while 6 systems are undefined due to insufficient sampling within the inner radial domain. The observed $\Delta\sigma \approx 0$ should therefore be interpreted as numerical neutrality of the adopted inner-region scatter metric under bounded local perturbations, rather than as the absence of local profile response, which is captured independently by the nonzero χ–ΔV correlations. Formally,

$$\Delta\sigma = \sigma_{\rm H2} - \sigma_{\rm H1} \approx 0\,{\rm km\,s^{-1}}, \tag{12}$$

where σ is defined in Eq. (11).

Numerically, $|\Delta\sigma| < 10^{-12}$ km s^{-1} across all 74 measurable systems, consistent with machine-precision floating-point rounding. Across all 80 galaxies, the numerical consistency check was never flagged.

Within numerical precision, the adaptive scale modulation does not produce a measurable change in the inner-region scatter metric relative to H1. The residual distributions of H1 and H2 are statistically indistinguishable under identical weighting and sampling.

4.2. Regime-dependent behavior

Although the inner scatter metric remains invariant, the adaptive mechanism modifies the local velocity profile in a regime-dependent manner.

Outer regime. In low-acceleration outer regions, where $\chi(r)$ remains small, $L_{\rm eff}(r)$ approaches the upper bound L_0, effectively recovering the frozen H1 configuration. No systematic deviation from the H1 outer-profile behavior is observed across the sample.

Inner regime. In baryon-dominated inner regions, $\chi(r)$ increases and $L_{\rm eff}(r)$ preferentially maps toward lower basis scales approaching $L_0/3$. The value $L_0/3$ should therefore be interpreted as an empirical operating floor of the validated sample under the realized adaptive response, not as the hard minimum of the implemented discrete basis. This produces localized adjustments in $V_{\rm NL}(r)$ relative to H1. However, the total velocity $V_{\rm tot}(r)$ remains dominated by the baryonic component under quadrature combination (Eq. 1), limiting the impact of nonlocal scale contraction on the overall residual amplitude.

4.3. Correlation analysis

To assess whether the adaptive mechanism responds coherently to baryonic structure, we compute the Pearson correlation coefficient between $\chi(r)$ and $\Delta V(r)$ across the evaluated radial domain. We find systematic variation in correlation strength across dynamical regimes:

- Baryon-dominated systems (39 galaxies, $V_{\rm obs}/V_b < 0.8$): $r_{\rm p}$ ranges from −0.371 to +0.981, with median $r = +0.54$ (bootstrap 95% CI: [+0.46, +0.76])
- Balanced transition regimes (30 galaxies, $0.8 \le V_{\rm obs}/V_b < 1.5$): $r_{\rm p}$ ranges from −0.169 to +0.953, with median $r = +0.75$ (bootstrap 95% CI: [+0.57, +0.88])
- Dark-matter-dominated systems (11 galaxies, $V_{\rm obs}/V_b \ge 1.5$): $r_{\rm p}$ ranges from +0.476 to +0.911, with median $r = +0.71$ (bootstrap 95% CI: [+0.58, +0.88])

The peak correlation across the sample (UGC07577, $r_{\rm p} = +0.981$) confirms that the modulation field operates coherently within the bounded scale interval. The systematic trend across regimes supports a regime-dependent adaptive response rather than uniform behavior.

4.4. Magnitude scaling

The maximum absolute velocity correction scales monotonically with regime:

- Baryon-dominated: $\max|\Delta V| \le 0.43$ km s^{-1}
- Balanced: $\max|\Delta V| \approx 0.15$–$0.40$ km s^{-1}
- Dark-matter-dominated: $\max|\Delta V| \approx 2.5$ km s^{-1}

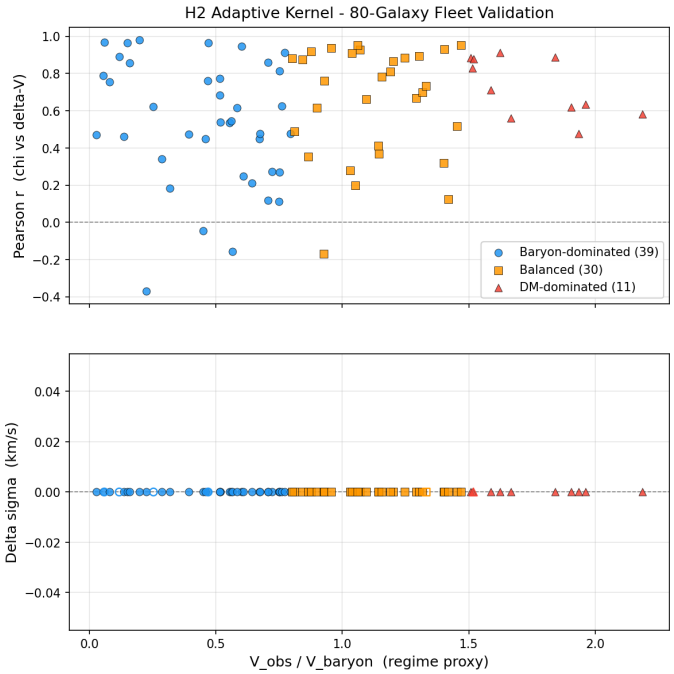

Fig. 1. Diagnostic metrics across dynamical regimes for the 80-galaxy sample. Top: Pearson correlation r_p between χ and ΔV. Bottom: Change in scatter $\Delta\sigma$ (km s^{-1}). For 74 measured galaxies, $\Delta\sigma \approx 0$ km s^{-1}; 6 galaxies with insufficient inner-region coverage are shown as open symbols. Correlation strength varies systematically with $V_{\rm obs}/V_b$ across all three dynamical regimes.

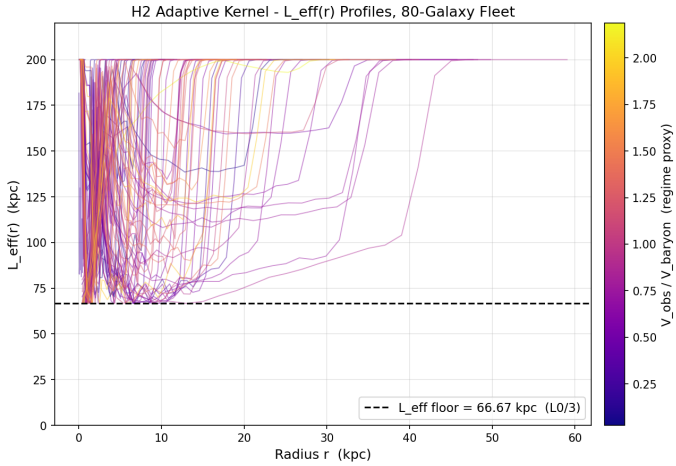

Fig. 2. Adaptive kernel scale $L_{\rm eff}(r)$ for all 80 galaxies (colored by $V_{\rm obs}/V_b$ ratio). All profiles reach or approach the empirical operating floor $L_0/3 = 66.67$ kpc (dashed horizontal line) in high-stiffness inner regions, independent of regime classification ($L_{\rm eff,min}$ range: 66.67–96.95 kpc).

Table 1. Summary statistics for the 80-galaxy H2 validation sample.

Parameter	Value
Total galaxies	80
Baryon-dominated ($V_{\rm obs}/V_b < 0.8$)	39
Balanced ($0.8 \le V_{\rm obs}/V_b < 1.5$)	30
DM-dominated ($V_{\rm obs}/V_b \ge 1.5$)	11
$V_{\rm obs}/V_b$ range	0.027–2.187
Pearson r range	−0.371–+0.981
$\Delta\sigma \approx 0$ km s^{-1} (measured)	74
$\Delta\sigma$ = N/A (inner coverage)	6
Diagnostic protocol violations	0
$L_{\rm eff}$ floor (fleet-wide)	66.67 kpc
$L_{\rm eff,min}$ range	66.67–96.95 kpc

Notes. $V_{\rm obs}/V_b$: observational velocity ratio used as a proxy for dynamical regime, computed as the ratio of peak velocity amplitudes of each component across the galactic radial profile. $\Delta\sigma$: change in inner scatter relative to H1 baseline. For all 74 measurable galaxies, $\Delta\sigma$ is consistent with zero within floating-point tolerance ($|\Delta\sigma| < 10^{-12}$ km s^{-1}). The 6 N/A cases arise from data geometry (insufficient inner radial coverage) rather than model failure. Complete per-galaxy results are provided in the archived repository as `fleet_summary_80galaxy.csv`.

These corrections, while coherent with the adaptive mechanism, remain 1–2 orders of magnitude below typical rotation curve residuals (∼10–50 km s^{-1}) and observational uncertainties.

4.5. Geometric floor identification

Across the 80-galaxy validation sample, the empirical operating floor remains fixed at $L_{\rm eff,min} \approx L_0/3 = 66.67$ kpc, independent of regime, morphology, or χ spatial pattern. Many systems reach this floor directly, while others approach it asymptotically ($L_{\rm eff,min}$ range: 66.67 to 96.95 kpc), an operational limit within the present implementation. This floor reflects the realized χ-field range and taper constraints of the adaptive response under fixed $L_0 = 200$ kpc, and defines a structural boundary within the validated sample.

5. Discussion

5.1. Quadrature suppression mechanism

The invariance of the inner scatter statistic ($\Delta\sigma \approx 0$ km s^{-1} for all measurable systems) indicates that bounded local scale modulation does not alter the inner-region scatter of the model under quadrature combination (Eq. 1). This behavior arises directly from the mathematical structure of the total velocity. This result also clarifies the trajectory from H1.5 to H2. The H1.5 diagnostics identified structured local mismatch under frozen canonical constraints, while H2 tests whether bounded local scale adaptation can remove that mismatch without altering the baseline implementation. The present 80-galaxy analysis shows that this form of bounded local adaptation does not alter the inner scatter metric within the validated subset. In baryon-dominated inner regions, the Newtonian component $V_{\rm b}$ contributes the dominant term to $V_{\rm tot} = \sqrt{V_{\rm b}^2 + V_{\rm NL}^2}$. In this regime, variations in $V_{\rm NL}$ induced by scale contraction produce second-order corrections to the total velocity. The sensitivity of $V_{\rm tot}$ to changes in $L_{\rm eff}$ scales approximately as

$$\frac{\partial V_{\rm tot}}{\partial V_{\rm NL}} = \frac{V_{\rm NL}}{\sqrt{V_b^2 + V_{\rm NL}^2}} \tag{13}$$

When $V_{\rm b} \gg V_{\rm NL}$, this sensitivity approaches zero, effectively suppressing the impact of adaptive kernel modulation on the total velocity profile. Consequently, even strong χ–ΔV correlations (e.g., $r_{\rm p} = +0.95$ in IC2574, $r_{\rm p} = +0.981$ in UGC07577) do not translate into measurable reductions in the total residual amplitude. Within the present validation sample, the suppression identified in Eq. 13 is most effective when the nonlocal contribution remains subdominant relative to the baryonic term in the inner regions. The fixed scale $L_0 = 200$ kpc should therefore be inter-

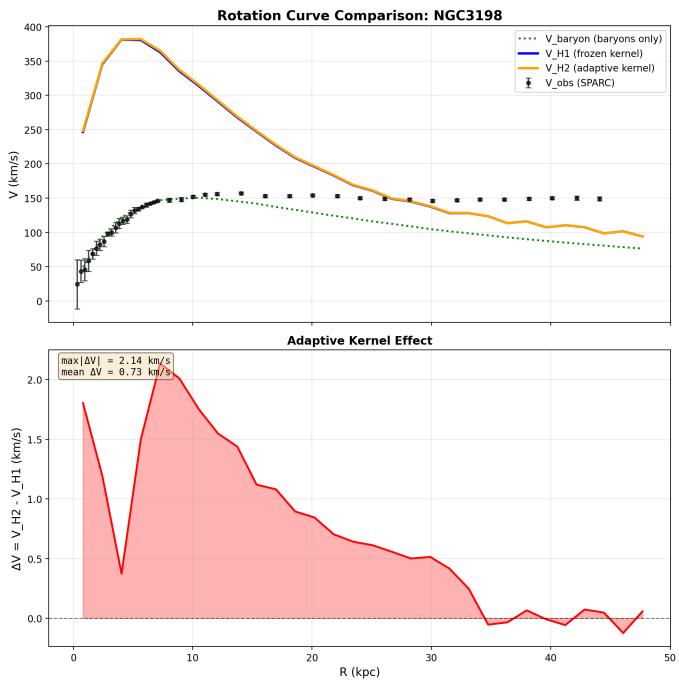

Fig. 3. Rotation-curve comparison for NGC3198 (balanced regime, $V_{\rm obs}/V_b = 1.42$). Observed data (points), H1 baseline (solid line), and H2 adaptive model (dashed line). Despite inner $L_{\rm eff}$ contraction toward the empirical operating floor, quadrature suppression limits the impact on the total velocity profile.

preted here as a phenomenological coupling scale of the implementation rather than as a localized interaction range tied to optical disk size. This relation shows that perturbations in the nonlocal component are suppressed when $V_{\rm NL} \ll V_b$, explaining why adaptive scale modulation produces limited changes in the total velocity. This suppression arises from the quadrature structure itself, rather than from numerical instability or insufficient correlation strength. The adaptive mechanism operates coherently within its allowed bounds, but the mathematical coupling limits its leverage in regions where one component dominates.

5.2. Role of the empirical operating floor

The adaptive response saturates at an empirical operating floor $L_{\rm eff} \approx L_0/3$, which is reached systematically in high-stiffness regions across the sample. This floor emerges from the realized χ-field range and taper constraints rather than from the hard minimum of the discretized basis (which extends to $0.05\,L_0$), and it prevents further contraction of the nonlocal kernel in practice.

The fact that the adaptive mechanism saturates at this floor without modifying the inner scatter metric indicates that further reduction of inner residuals cannot be achieved through scale contraction alone while L_0 and μ remain fixed. The empirical floor therefore represents an operational boundary of the present framework: within fixed global parameters, local scale adaptation is insufficient to modify the overall statistical dispersion of the rotation curve residuals.

5.3. Assessment of achieved and unachieved objectives

The H2 extension shows:

- coherent response of the kernel scale to baryonic gradient structure,
- regime-dependent modulation of $V_{\rm NL}$,
- preservation of outer-regime stability,
- strict numerical neutrality in the inner scatter metric.

However, the adaptive extension does not:

- reduce the inner-region scatter metric,
- eliminate inner-region overshoot,
- introduce measurable improvement in rotation-curve fits.

These findings indicate that the residual structure observed in H1 is not attributable solely to the absence of local scale modulation. Instead, it appears to be a structural property of the isotropic kernel formulation under fixed global parameters.

5.4. Interpretation of scatter neutrality

The consistent result of $\Delta\sigma \approx 0\,{\rm km\,s^{-1}}$ for all galaxies with sufficient inner-region coverage, together with 6 undefined inner-coverage cases, warrants careful interpretation. Several independent lines of evidence support the interpretation that this is a systematic pattern rather than a numerical artifact.

(1) Active kernel response. Pearson correlations range from −0.371 to +0.981, indicating the adaptive kernel responds differently to each galaxy's local baryonic structure. The modulation field is not dormant.

(2) Independence from baseline quality. Systems with poor H1 baseline fits (MAFE > 0.9, including NGC2403 with MAFE = 1.67 and NGC0024 with MAFE = 1.57) maintain scatter neutrality identically to those with excellent fits (MAFE < 0.1), confirming that the result is not sensitive to H1 fit quality across the tested range.

(3) Fleet-wide empirical floor. All 80 galaxies reach or approach $L_{\rm eff,min} \approx L_0/3 = 66.67$ kpc, confirming that the adaptive mechanism reaches its empirical operating floor across the full range of dynamical regimes.

(4) Sample breadth. 80 galaxies spanning $V_{\rm obs}/V_b = 0.027$ to 2.187 constitute a broader validation sample across morphological types, surface brightness regimes, and baryonic fractions.

The observed pattern — active local adaptation with invariant scatter — could reflect: (a) a structural constraint arising from the quadrature velocity coupling, (b) saturation at the geometric floor preventing further kernel contraction, or (c) limited sensitivity of the adopted scatter metric to sub-km s^{-1} local adjustments. Distinguishing these interpretations requires further investigation beyond the scope of this diagnostic study.

5.5. Implications for bounded nonlocal models

The present analysis defines a constraint on state-driven adaptive nonlocal kernels operating under quadrature combination. When the effective contribution of the nonlocal term remains subdominant, bounded local modulation does not alter the inner scatter statistic.

This result provides a diagnostic benchmark for future extensions of nonlocal modeling frameworks. Within the present framework, any modification intended to reduce the inner scatter metric must overcome the suppression mechanism identified here, either through alternative coupling structures, modified kernel functional forms, or adjustments to global baseline parameters.

The H2 results therefore serve as a controlled null-response baseline for evaluating more complex formulations. The framework establishes reproducible boundary conditions that can inform the design space of adaptive gravitational models.

6. Conclusions

We have characterized the structural and statistical limits of a bounded adaptive nonlocal kernel framework (H2) for modeling galaxy rotation curves. By extending a frozen baseline configuration with a state-driven scale-modulation field $\chi(r)$, we isolated the impact of local baryonic structure on the effective gravitational response. Diagnostic evaluation across an 80-galaxy validation sample from the SPARC database reveals three principal findings.

First, the adaptive mechanism demonstrates coherent regime-dependent coupling between the stiffness field $\chi(r)$ and local velocity adjustments. The systematic variation in Pearson correlation from $r_{\mathrm{p}} = -0.371$ to $+0.981$ across dynamical regimes indicates that scale modulation operates consistently within the tested implementation.

Second, the empirical operating floor $L_{\mathrm{eff}} \approx L_0/3$ is reached or approached in high-stiffness regions across all 80 galaxies ($L_{\mathrm{eff,min}}$ range: 66.67–96.95 kpc). This floor establishes an operational boundary for local adaptation under fixed global scaling and prevents further contraction of the nonlocal kernel within the validated sample.

Third, and most significantly, $\Delta\sigma$ is consistent with zero for all galaxies with sufficient inner-region coverage; 6 systems are undefined due to inner-coverage limitations. Across all 80 galaxies, no measurable departure from inner-scatter neutrality was found beyond floating-point tolerance. Quadrature coupling, together with the empirical operating floor and the bounded adaptive range, suppresses the influence of local scale contraction on the total velocity, and local state-driven deformation alone is insufficient to modify the inner-region scatter metric.

These findings indicate that the residual structure observed in the frozen baseline cannot be removed solely by introducing bounded local scale variability. Within the bounded adaptive implementation examined here, modifications that aim to alter the inner scatter metric must overcome the suppression mechanism identified in this study.

The H2 implementation is publicly archived and fully reproducible. It provides a reproducible diagnostic benchmark for bounded adaptive nonlocal kernel models.

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Data availability

The complete H2 framework code, diagnostic pipeline, and 80-galaxy validation results, including `fleet_summary_80galaxy.csv` and associated figures from `fleet_expansion_80galaxy/`, are publicly archived at `https://doi.org/10.5281/zenodo.18793233` under an MIT license. The development repository for the project is available at `https://github.com/VitorFigueiredoResearch/H2-Adaptive-Kernel-Framework-for-Galaxy-Rotation-Curves`. Rotation curve data from the SPARC database (Lelli et al. 2016) are available at `http://astroweb.cwru.edu/SPARC/`. All results presented in this paper are reproducible using the archived code together with the publicly available SPARC data.

Appendix A: H1 Baseline Framework: Formalism and Implementation

The H2 adaptive framework builds upon the H1 frozen baseline (Figueiredo 2025a,b). To ensure reproducibility and self-containment, we provide both (1) the formal H1 equations as presented in the manuscript, and (2) the specific numerical implementation used in this work, archived publicly at https://doi.org/10.5281/zenodo.18793233.

A.1. H1 formalism (from manuscript)

The H1 effective potential is decomposed as:

$$\Phi_{\rm eff}(\mathbf{r}) = \Phi_{\rm b}(\mathbf{r}) + \Phi_{\rm K}(\mathbf{r}), \tag{A.1}$$

where $\Phi_{\rm b}$ is the Newtonian baryonic potential and $\Phi_{\rm K}$ is the kernel-induced nonlocal contribution defined by:

$$\Phi_{\rm K}(\mathbf{r}) = \mu G \int \rho_{\rm b}(\mathbf{r}')\, U(|\mathbf{r}-\mathbf{r}'|; L)\, d^3r', \tag{A.2}$$

where μ is a dimensionless coupling parameter, L is the kernel characteristic scale, and $U(r; L)$ is an isotropic radial kernel function normalized such that $\int U(r; L) d^3r = 1/L$.

The circular velocity is derived from the radial gradient of the effective potential:

$$V_{\rm tot}^2(R) = R \left.\frac{\partial \Phi_{\rm eff}}{\partial R}\right|_{z=0}. \tag{A.3}$$

For numerical stability, this is implemented as a quadrature decomposition:

$$V_{\rm tot}^2(R) = V_{\rm b}^2(R) + V_{\rm K}^2(R), \tag{A.4}$$

where $V_{\rm b}$ is the Newtonian baryonic contribution and $V_{\rm K}$ is the kernel-induced contribution. This decomposition is a computational identity and does not imply physical independence (Figueiredo 2025a).

The H1 framework explores a fixed parameter grid:

$$L \in [20, 200]\,{\rm kpc}, \quad \mu \in [1, 500], \tag{A.5}$$

sampled logarithmically to identify optimal (L, μ) pairs for each galaxy.

A.2. Numerical implementation specification

To ensure exact reproducibility of the H2 diagnostic pipeline, we specify the numerical implementation of H1 used in this work, publicly archived with DOI 10.5281/zenodo.18793233.

A.2.1. Kernel functional form

While the H1 manuscript defines the kernel $U(r; L)$ via its normalization and asymptotic properties, the specific analytic form implemented in the diagnostic pipeline is a regularized logarithmic kernel:[1]

$$U(r; L) = \frac{1}{2} \ln\left(1 + \frac{r_{\rm safe}^2}{L^2}\right), \tag{A.6}$$

where $r_{\rm safe} = \sqrt{r^2 + \epsilon^2}$ with $\epsilon = 0.01\,L$ prevents singularity at $r = 0$. This functional form is consistent with the asymptotic properties described in H1 while providing numerical stability at small radii.

A finite-support taper is applied:

$$T(r) = \begin{cases} 1, & r \le L \\ \exp\left[-\left(\frac{r-L}{\Delta}\right)^2\right], & r > L, \end{cases} \tag{A.7}$$

with $\Delta = L$, truncating the kernel at $R_{\rm cut} = 6L$.

A.2.2. Amplitude calibration

Following validation diagnostics performed in H1.5, a global amplitude correction factor $\beta = 1.15$ was identified as necessary to account for finite-grid sampling effects and the discrete vertical disk prescription. This constant is applied uniformly across all galaxies in the archived implementation:

$$\hat{\Phi}_{\rm K}(\mathbf{k}) = \mu\beta G\, \hat{\rho}_{\rm b}(\mathbf{k})\, \hat{U}(\mathbf{k}), \tag{A.8}$$

where the convolution is evaluated in Fourier space for computational efficiency.

A.3. Eighty-galaxy H2 baseline parameters

All 80 galaxies in the H2 validation set share identical H1 frozen parameters in the archived implementation:

$$L_0 = 200\,{\rm kpc}, \quad \mu = 10, \quad \beta = 1.15. \tag{A.9}$$

These values define the compatibility condition for H2 adaptive convolution under taper constraints. Galaxies whose H1 best-fit parameters differ from this canonical baseline require recalibration of the adaptive response and were excluded to maintain the homogeneity of the present diagnostic study (see Section 3.1).

Table A.1 presents representative H1 baseline rotation-curve metrics for the original 30-galaxy validation subset used during the staged construction of the 80-galaxy H2 sample.

The H2 adaptive framework retains this H1 baseline in its entirety across all 80 galaxies, introducing only local kernel scale modulation $L_{\rm eff}(r)$ as a diagnostic modification while holding (L_0, μ, β) fixed. All H2 results are differential comparisons against this frozen baseline.

[1] This implementation corresponds to the kernel architecture archived in the project repository and referred to internally in the development repository as the hybrid kernel.

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Table A.1. Representative H1 baseline rotation-curve metrics for the original 30-galaxy validation subset of the 80-galaxy H2 sample. All values computed from the archived H1 frozen-parameter pipeline with $(L_0, \mu, \beta) = (200\,\mathrm{kpc}, 10, 1.15)$.

Galaxy	$\mathrm{MAFE}_{\mathrm{H1}}$	$V_{\mathrm{kernel}}/V_{\mathrm{bar}}$
NGC 6946	0.139	1.11
NGC 2903	0.227	1.73
NGC 5585	0.409	0.95
NGC 0891	0.120	0.92
NGC 7793	0.380	0.73
NGC 2998	0.141	1.23
NGC 0024	1.569	2.94
IC 2574	0.224	0.81
NGC 3893	0.188	1.42
DDO 161	0.478	1.13
NGC 3769	0.759	2.38
NGC 3726	0.510	2.03
NGC 1003	0.277	1.29
NGC 3198	1.380	2.40
NGC 5033	0.304	2.37
NGC 3953	0.088	1.21
NGC 3521	0.429	1.51
NGC 3992	0.258	2.57
UGC 03580	1.398	2.04
UGC 06983	0.124	1.43
NGC 5055	0.387	2.26
UGC 07608	0.190	1.16
UGC 02259	0.226	2.14
NGC 4100	0.127	1.90
KK 98-251	0.217	1.15
UGC 05716	0.246	1.84
F 571-8	0.338	2.12
NGC 6503	0.226	2.75
NGC 2403	1.673	3.17
NGC 2955	0.175	2.25

Notes. Mean absolute fractional error (MAFE) quantifies rotation curve fit quality. The ratio V_K/V_b reported here is an H1 baseline diagnostic quantity and is distinct from the V_{obs}/V_b regime proxy used in the main H2 analysis. This table shows the original 30-galaxy validation subset. The complete 80-galaxy baseline parameters are archived as `fleet_summary_80galaxy.csv`.